Roberto Trotta

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Main Academic Appointments

Current

2024– **Director, Interdisciplinary Lab (ILAS)**
International School for Advanced Studies (SISSA), Trieste, Italy

2023– **Full Professor of Theoretical Physics (Professore di I fascia)**
Physics Department, SISSA, Trieste, Italy

Previous Academic Positions

2021–25 **Head, Theoretical and Scientific Data Science group**
Physics Department, SISSA, Trieste, Italy

2019–22 **Full Professor of Astrostatistics (on leave 2020–22)**
Physics Department, Imperial College London, UK

2015–20 **Director, Centre for Languages, Culture and Communication**
Education Office, Imperial College London, UK

2008–19 **Lecturer (2008–12), Senior Lecturer (2012–16), Reader (2016–19)**
Physics Department, Imperial College London, UK

2006–08 **Junior Fellow**
St Anne's College, University of Oxford, UK

2005–08 **Sir Norman Lockyer Research Fellow of the Royal Astronomical Society**
Astrophysics Department, University of Oxford, UK

2004–05 **Postdoctoral Researcher**
Department of Theoretical Physics, University of Geneva, Switzerland

Education and Qualifications

2022, 2020 **Abilitazione Scientifica Nazionale a professore di I fascia**
Settori FIS 02/A2 (valid until 2032) and FIS 02/C1 (valid until 2029)

2011 **Certificate of advanced study in learning and teaching**
Centre for Higher Education Research and Scholarship, Imperial College London

2004 **PhD in Theoretical Physics**
University of Geneva, Switzerland. Supervisor: Ruth Durrer

2001 **MSc(Hons) in Physics**
Swiss Federal Institute of Technology (ETHZ), Zurich, Switzerland

Selected Prizes and Awards

Annie Maunder Medal for Outreach, Royal Astronomical Society (2020) — the highest RAS recognition for public engagement; **President's Award for Leadership in Public Engagement**, Imperial College London (2018); **Chaire Georges Lemaître** (2018), Université Catholique de Louvain; **Royal Society Significance Lecture** (2017); **Outstanding Performance Award**, Imperial College (2017); **President's Award for Excellence in Teaching**, Imperial College London (2016); **Foreign Policy 100 Global Thinkers** (2014); **Michelson Postdoctoral Lectureship Prize**, Case Western Reserve University (2008); **Lord Kelvin Award Lecture**, British Association for the Advancement of Science (2007).

Visiting Academic Positions

2023– **Visiting Professor of Astrostatistics**
Imperial College London, UK

2023– **Visiting Faculty, Data-Driven Management**
MIB Trieste School of Management, Italy

2019–22 **Visiting Professor of Cosmology**
Gresham College, London, UK

2011–2013 **Next Einstein Visiting Scholar**
African Institute for Mathematical Sciences (AIMS), Cape Town, South Africa

Teaching and PhD Supervision

Teaching at SISSA (2020–). Bayesian Methods I & II; Bayesian Inference and Machine Learning in Cosmology; Ethical Aspects of AI (Theoretical and Scientific Data Science PhD); AI and Information (Master in Science Communication); Machine Learning for Business Analytics (MBA, MIB Trieste).

Teaching at Imperial College London (2008–2020). Lecturer of 'Physics of the Universe' (3rd-year core) and 'Statistics of Measurement' (2nd-year core); Cosmology (associate lecturer); 30+ international postgraduate schools and masterclasses (Saas-Fee, ICTP, Cargèse, Heidelberg, Niels Bohr Institute, AIMS, ESAC).

PhD Supervision. 9 completed PhD theses (incl. Springer Thesis Prize, two Winton Capital Best PhD Thesis prizes, RSS Emerging Applications Thesis Prize 2025); 2 ongoing PhDs; 14 postdocs (alumni now faculty at Cambridge, Uppsala, CONICET, Lawrence Livermore); 13 MSc/MSci students; 35 BSc projects; 18 external PhD/MSc examinations internationally.

National and International Collaborations and Networks

Experimental Collaborations. LSST Dark Energy Science Collaboration, member (2025–); XLZD Consortium for direct dark matter detection, member (2022–); DARWIN dark matter experiment, Executive Board member (2015–) and Science Working Package leader (2014–); ATLAS Collaboration, Short Term Associate (2014–16); XENON100, Associated Scientist (2014–18); Fermi-LAT, Affiliated Scientist (2012–); Euclid ESA mission, Theory Working Group member (2010–), likelihood Work Package co-lead (2020–22).

Networks and Consortia. Centro Nazionale FAIR (PNRR), Spoke Edge and Exascale Computing (2022–); Centro Nazionale HPC (PNRR), Spoke COSMOS (2022–).

Advisory and Coordination. Scientific Consultant, ICTP Trieste (2022); Co-lead, Astrostatistics section of EuCAPT White Paper (2021); Astrostatistics section lead, Euclid Theory Working Group (2013, 2018); Scientific Advisor on Dark Energy probes, STFC UK (2005–07).

Other Work Experience and Consultancy

Advisory Roles. Board of Directors, Fondazione Internazionale Trieste per il Progresso e la Libertà delle Scienze (2026–); Scientific Advisory Board, RACHAEL S.r.l. (2024–; previously Chairman of SAB 2021–24).

Entrepreneurship. Co-founder and Director, *Data Fusion Consultants Ltd* (2012–20); Non-executive Director, Imperial Consultants (2018–20).

Industry and Media Clients (Selection). Beluga Group, Savio Macchine Tessili, Oxford University Press, Nature Publishing Group, Red Planet Pictures (BBC 'Death in Paradise'), Left Bank Pictures (Netflix 'Origin'), Fantastic Productions (Netflix), BBC Sky at Night, London Science Museum, Deutsches Museum Munich, Museo del Castello di Miramare, Regione Friuli Venezia Giulia, Institute of Physics. Statistical consultancy for clients in shipping, energy, property, banking, fashion and online commerce.

Executive Training. Imperial College Business School Executive Education (Sberbank Digital Technology, 2020–21; Naspers Executives Programme, 2019); Generali Business Translators (2022); CINECA Academy (2022); Intesa San Paolo (2024); FINECO Advisory Seminar keynote (2025); ML for Business Analytics, MBA, MIB Trieste (2023–).

Administrative Roles and Responsibilities

SISSA, Trieste. Director, Interdisciplinary Lab (ILAS, 2025–); Head of Data Science (2021–25); PhD Coordinator and Academic Board Chair, Theoretical and Scientific Data Science (2021–25); Head of Group, Theoretical and Scientific Data Science (2021–25); Member, Physics Area Steering Board (2021–25); Knowledge Transfer Committee representative (2021–); Physics Area coordinator for the 2014–19 national assessment exercise (VQR, 2021).

Imperial College London. Director, Centre for Languages, Culture and Communication (CLCC, 2015–20); member of Imperial College Senate (2016–20); Education Strategy and Operations Group (2020); Education Office COVID-19 Response Group (2020); Horizons Stream Lead, I-Explore programme (2019–20); Chair, Curriculum Review Reference Panel (2018–19); Chair of CLCC Departmental Teaching Committee, CLCC Management Team, Imperial Horizons Team, and President's Award review panel (all 2015–20); co-chair, CLCC/CHERS Education Committee (2016–20).

Gresham College, London. Member of Academic Board (2019–22).

International. Member, EuCAIF Steering Board, European Coalition for AI in Fundamental Physics (2024–).

Scientific Organisations and Coordination of Academic Activities

International Executive and Steering Boards. International Executive Committee, INSAP conference series (2017–); Council of the International Astrostatistics Association (2015–); Outreach and Knowledge Transfer Leader, IAA (2015–); Astrostatistics Committee, International Statistical Institute (2011–); Working Group on Equity and Inclusion, Commission C1, IAU (2016–).

Conference and School Organisation (Selection from 40+ Events). LOC Chair, Convegno Nazionale di Comunicazione della Scienza, Trieste (2025); SOC, EuCAIFCon (Amsterdam 2024; Cagliari 2025); SOC, INSAP XI–XIII (CalTech 2022, Corfu 2024, Belfast 2025); SOC, Advanced Schools on Applied Machine Learning, ICTP (2024, 2026); SOC, Advanced School on Foundation Models for Scientific Discovery, ICTP (2025); Chair and founder, 'Cosmostat' conference series (Switzerland 2009; Banff 2013); Lead Organiser, KITP programme 'Multi-probe Identification of Dark Matter', UCSB (2013).

Evaluation and Assessment Panels. Evaluator: Agence Nationale de la Recherche, France (2026); Swiss National Science Foundation (2024, 2025); ERC Consolidator Grants (2026); ERC Synergy Grants (2025); ERC Starting Grants (2021); UKRI Future Leaders (2025); Dutch Research Council (2024); Canada Research Chairs programme (2020), South African National Research Foundation (2007–). Promotion and tenure assessor: Cambridge University (2025); U. Minnesota (2023). Panel member: STFC Rutherford Fellowships (2014–15); Royal Society International Grants (2007–12); Dutch Research Council ORC programme (2024, 2026).

Editorial Activity

Editorial Boards. Scientific Editor and founding member of the inaugural editorial board of *RAS Techniques and Instruments* (RASTI), open-access journal of the Royal Astronomical Society (2021–). Member, *Astronomy and Geophysics* Editorial Board (2009–).

Guest Editor. *Statistical Analysis and Data Mining* (Wiley), special issue on Astrostatistics with A. Gandy, vol. 6, 1–2 (2013); special issue, *Physics of the Dark Universe* (Elsevier, 2013).

Peer Review. 80+ papers refereed for *Nature Physics*, *Physical Review Letters*, *Physical Review D/E*, *ApJ*, *MNRAS*, *JCAP*, *JHEP*, *Physics Letters B*, *New Astronomy*, *Computer Physics Communications*, *Science and Education*, *RASTI*.

Membership of Scientific Societies

Elected Fellow. International Astrostatistics Association (2016); Royal Astronomical Society (FRAS); Royal Astronomical Society Dining Club (2012).

Academic / Senior Fellow. Imperial College London Data Science Institute (2018–2022); Higher Education Academy, UK (SFHEA, 2021).

Professional Member. International Society for Bayesian Analysis; International Astronomical Union; Swiss Society for Astronomy and Astrophysics; European Astronomical Society; International Statistical Institute — Astrostatistics Committee; Gresham Society.

Affiliate. Chartered Management Institute, UK (2021–).

Jury and Awards Panels. Jury Member, Premio Cosmos (Italian national science book award, 2022–); International FameLab semi-final judge, Cheltenham (2015).

Public Engagement and Science Communication

Books for the General Public. *Starborn: How the Stars Made Us* (Basic Books, 2023): BBC Radio 4 Book of the Week, BBC Sky at Night 5-star review, *New Scientist* '22 Best Non-Fiction and Popular Science Books of 2023', *Smithsonian Magazine* 'Ten Best Science Books of 2023', 2025 Nautilus Book Awards Gold Winner; translated into Italian, Korean, Spanish and Chinese. *The Edge of the Sky* (Basic Books, 2014): cosmology in the 1,000 most common English words; for which I was named *Foreign Policy* 'Global Thinker 2014'; translated into Korean, German and Catalan.

Public Reach. 198+ public lectures and talks at major science festivals (Cheltenham, Edinburgh, New Scientist Live, Royal Albert Hall, Royal Institution, Mantova, Torino, Genova); 47+ articles for the general public (*TIME*, *The Guardian*, *New Scientist*, *La Stampa*, *Le Scienze*); broadcast appearances on BBC Radio 4, BBC World Service, NPR, RAI, RSI Swiss National Radio/TV, Sky TG24.

Art and Science Projects. Co-author and creative director, *LIBRA* (2021, multimedia theatre play, performed at Miramare Castle, Reggio Calabria, Gorizia 2025 European Capital of Culture); AI consultant, *Parl-IA-moci* (2023, theatre play with Diana Höbel and ChatGPT); *A Multi-Sensorial Dark Matter Experience* (2019); *Dinner with Copernicus* (2018); *g-ASTRONOMY* multi-sensorial cosmology dinners; *The Theory of Everything* short film (audience award, International Short Film Festival Berlin 2013).

Consulting for Film and Media. Scientific consultant for Red Planet Pictures (BBC 'Death in Paradise'), Left Bank Pictures ('Origin', Netflix), Fantastic Productions (Netflix feature film), Nature Publishing Group, Oxford University Press.

Scientific Publications Record

Output. 115 papers in peer-reviewed physics journals or machine learning conferences (12 as first author, 3 in *Phys. Rev. Lett.*, 1 in *Nat. Astron*); 18 conference proceedings; 5 white papers and editorial works; 5 book chapters. The single-author review *'Bayes in the sky'* (Contemporary Physics, 2008) has 1500+ citations.

Citations (August 2026). 14,500+ citations, h-index 54 (INSPIRE/NASA ADS, higher of each per paper); 13,000+ citations, h-index 57 (Google Scholar). Citation thresholds: 22 papers >100 citations; 2 papers >500; 3 papers >1000. *Full publications list given in Part II.*

Major Community and White Papers. *'Opportunities in AI/ML for the Rubin LSST Dark Energy Science Collaboration'* White Paper (2026); *Nature Astronomy* opinion piece (2025): 'The indiscriminate adoption of AI threatens the foundations of academia'; co-lead of the Astrostatistics section, EuCAPT White Paper (2021); Astrostatistics section lead, Euclid Theory Working Group *'Cosmology and Fundamental Physics with the Euclid Satellite'*, Living Reviews in Relativity (2013, 2018).

Competitive Grants Record

Full list and per-section breakdown in Part II. Figures below are the running total across all competitive grants held as PI or co-I (excludes the one declined award).
Overall total: *£10.4M • €2M*

Part II — Detailed Record

Full lists of appointments, details on grants, publications, knowledge transfer activities and public engagement.

ACADEMIA

Academic Appointments (full list)

2024–	Director, Interdisciplinary Lab (ILAS), SISSA, Trieste
2023–	Professor of Theoretical Physics (Professore di I fascia), Physics Department, SISSA, Trieste
2021–25	Head of Data Science, SISSA, Trieste
2020–22	Associate Professor (Professore di II fascia), Physics Department, SISSA, Trieste
2019–22	Professor of Astrostatistics, Imperial College London
2015–20	Director, Centre for Languages, Culture and Communication, Imperial College London
2016–19	Reader in Astrophysics (equivalent to Associate Professor), Imperial College London
2013–17	Public Engagement Fellowship, Science and Technology Facilities Council, UK
2012–16	Senior Lecturer in Astrophysics, Imperial College London
2008–12	Lecturer in Astrophysics (equivalent to Assistant Professor), Imperial College London
2006–08	Junior Fellow, St Anne's College, Oxford
2006	Junior Fellow Linacre College, Oxford (declined)
2005–08	Sir Norman Lockyer Research Fellow of the Royal Astronomical Society, University of Oxford
2005	European Space Agency Research Fellowship (declined)
2004–05	Research Associate, University of Geneva, Switzerland

Honours and Awards

2022	Charmian Brinson Honorary Lecture, Imperial College London
2020	Annie Maunder Medal for Outreach, Royal Astronomical Society
2018	Chaire Georges Lemaître, Université Catholique de Louvain, Belgium; President's Award for Leadership in Public Engagement, Imperial College London
2017	Significance Lecture, Royal Statistical Society; Outstanding Performance Award, Imperial College London; Highly commended, President's Award for Excellence in Societal Engagement, Imperial College London
2016	Longlisted, NCCPE Engage Competition Award (individual project category); President's Award for Excellence in Teaching, Imperial College London; Faculty of Natural Sciences Prize for Excellence in Teaching, Imperial College London
2015	Excellence Award, Stio town hall council, Salerno, Italy
2014	Foreign Policy 100 Global Thinkers
2012	Elected to the Royal Astronomical Society Dining Club
2011	Merit Award, Imperial College London
2008	Michelson Postdoctoral Lectureship Prize, Case Western Reserve University, USA
2007	Lord Kelvin Award Lecture, British Association for the Advancement of Science; Merit Award, Royal Astronomical Society; Merit Award, Oxford University

Distinguished Fellowships and Professional Memberships

Elected Fellow. International Astrostatistics Association (2016); Royal Astronomical Society (FRAS); Royal Astronomical Society Dining Club (2012).

Academic / Senior Fellow. Imperial College London Data Science Institute (2018–); Higher Education Academy, UK (Senior Fellow, 2021; Fellow status achieved 2011).

Professional Member. International Society for Bayesian Analysis; International Astronomical Union; Swiss Society for Astronomy and Astrophysics; European Astronomical Society; Gresham Society.

Affiliate. Chartered Management Institute, UK (2021–2022).

Continuing Professional Development

2015–21	Imperial Leadership and Management Development Programme (validated by the Chartered Management Institute, UK). Courses on: introduction to management; managing groups and teams; leading successful teams; effective recruitment and selection; finance for non-finance managers; promoting equality and diversity; CPD portfolio; 360 review and action planning; decision making; unconscious bias; managing difficult situations.
2016	Introduction to IP rights
2014	Bond Solon Expert Witness training programme

Scientific Editor

2021–	Scientific Editor and founding member of the inaugural editorial board of *RAS Techniques and Instruments* (RASTI), the open-access journal of the Royal Astronomical Society covering data collection methods, instrumentation, data processing and modelling, AI and machine learning, statistical methods, software, and high-performance computing.

Scientific Activities

2026–	Management Committee Member, EU Cost Action ‘Artificial Intelligence for Fundamental Physics: Open Roadmapping and Inclusive Actions (AIFORIA)’
2025–	Member, LSST Dark Energy Science Collaboration
2022–	Scientific Consultant, International Centre for Theoretical Physics (ICTP), Trieste; Programme Committee Member, ‘ML and AI for Science’, FVG Data Science Institute; Member, XLZD Consortium for direct dark matter detection
2014–	DARWIN direct dark matter detection experiment: Executive Board member (2015–), Science Working Package leader (2014–)
2014–16	ATLAS Collaboration, Short Term Associate
2014–18	XENON100 direct dark matter detection experiment, Associated Scientist
2012–	Fermi-LAT Affiliated Scientist
2012, 2013	Guest editor: *Statistical Analysis and Data Mining* special issue (2012); *Physics of the Dark Universe* special issue (2013)
2010–	Euclid ESA space mission, Theory Working Group member; likelihood analysis Work Package co-lead (2020–22)
2005–07	Scientific Advisor on Dark Energy probes, STFC, UK
2006–15	Co-author of the `SuperBayeS` public code for the analysis of supersymmetric theories
2001–	30+ invited scientific talks, 110+ research talks and seminars

Service

2026	Dutch Research Council evaluator, ERC Consolidator Grants evaluator
2025	Emmanuel College Cambridge Research Fellowship assessor; Cambridge University promotion assessor; UKRI Future Leaders evaluator; ERC Synergy Grants evaluator
2024	Dutch Research Council evaluator; Swiss National Science Foundation evaluator
2023	Tenure committee reviewer, School of Statistics, University of Minnesota
2021	ERC Starting Grants evaluator
2020	Referee, Canada Research Chairs programme
2018	Member, STFC Public Engagement Early Career Researcher Network Review Panel
2016–	Member, Working Group on Equity and Inclusion, Commission C1 of the IAU
2015–	Member, DARWIN-LXe Executive Board; Outreach and Knowledge Transfer Leader, International Astrostatistics Association; Member, Council of the International Astrostatistics Association
2015	University Research Fellowships referee, Royal Society; Imperial College London representative, Interdisciplinary Curriculum Group
2014–15	STFC Rutherford Fellowships panel
2011–	Member, Astrostatistics Committee, International Statistical Institute
2010	Founding member, Imperial Centre for Inference and Cosmology
2009	Jesus College and Peterhouse College fellowships referee, Cambridge
2009–2012	Member, *Astronomy & Geophysics* Editorial Board
2007–	South African National Research Foundation evaluator
2007–12	Royal Society International Grants Panel

Ongoing	Referee for *Nature Physics*, *PRL*, *ApJ*, *JCAP*, *JHEP*, *PLB*, *Phys. Rev. D/E*, *New Astron.*, *MNRAS*, *Comp. Phys. Comm.*, *Science and Education*, *RASTI* — 80+ papers refereed.

Institutional Responsibilities, Management and Leadership

SISSA, Trieste

2025–	Director, Interdisciplinary Lab (ILAS)
2021–25	PhD Coordinator and Academic Board Chair, Theoretical and Scientific Data Science; Head of Group, Theoretical and Scientific Data Science; Member, Giunta dell'Area Fisica (Physics Area Steering Board)
2021–	Member, Laboratorio Interdisciplinare per le Scienze Naturali (ILAS); Physics Area representative, Knowledge Transfer Committee (Commissione per la Valorizzazione del Talento e il Trasferimento della Conoscenza)
2021	Physics Area coordinator for the 2014–19 national assessment exercise (VQR)

Gresham College, London

2019–22	Member of the Academic Board

International

2024–	Member, EuCAIF Steering Board, European Coalition for AI in Fundamental Physics

Imperial College London

2020	Education Strategy and Operations Group
2019–20	Horizons Stream Lead, I-Explore programme
2020	Education Office COVID-19 Response Group
2018–19	Chair, Curriculum Review Reference Panel
2015–20	Chair, CLCC Departmental Teaching Committee
2016–20	Chair, President's Award review panel 'Supporting the Student Experience'
2015–20	Chair, Imperial Horizons Team; Chair, CLCC Management Team
2016–20	Co-chair, CLCC/CHERS Education Committee
2018–20	Member, I-Explore Modules Innovation Group; Member, Enterprise Academic Advisory Group; Member, Engagement Strategy Group
2015–20	Member, Programmes Committee
2013–20	Member, Imperial SpaceLab Steering Group
2016–20	Member, Strategy Board, Centre for Performance Science
2017–20	Member, Online Learning Innovation Group
2016–18	Member, Academic Partnerships Engagement with Research Steering Group
2015–20	Member, Education Office Management Committee
2016–20	Member, Imperial College Senate
2015–18	Member, Imperial Horizons Committee
2015–17	Member, Task and Finish Group for Assessment

Organising of Scientific Meetings

2026 SOC, 'DESC Virtual Collaboration Meeting' (online)
- ▹ LOC, 'Quasar Bazar', IFPU, Trieste
- ▹ SOC, Advanced School in Applied Machine Learning: Uncertainty and Robustness of DL Models for Science, ICTP
- ▹ SOC and co-proposer, NeurIPS workshop 'AI and Science: Evolution or Extinction?'

2025 LOC Chair, Convegno Nazionale di Comunicazione della Scienza, Trieste
- ▹ SOC, Advanced School on Foundation Models for Scientific Discovery, ICTP, Trieste
- ▹ SOC, AI & Physics track, 'AI4Science' conference, Ljubljana
- ▹ SOC, AI Alliance Meet-up, Trieste
- ▹ SOC, INSAP XIII conference, Belfast
- ▹ SOC, EuCAIFCon 2025, Cagliari

2024 SOC, Advanced School on Applied Machine Learning, ICTP, Trieste
- ▹ SOC, StatPhys conference, Imperial College London
- ▹ SOC, EuCAIFCon 2024, Amsterdam
- ▹ SOC, INSAP XII conference, Corfu
- ▹ LOC, 'Unsupervised Exploration of Fast Radio Bursts Dynamic Spectra', IFPU Team Research

Workshop, Trieste

- ▹ SOC, 'Statistics meets ML', Imperial College London
- ▹ SOC, COSMO21 conference, Chania

2023 Organiser, 'AI Interactions: ChatGPT and Beyond', Trieste and online

- ▹ SOC, 'Machine Learning Enabled Searches for New Physics in Astrophysical Data', IFPU, Trieste

2022 Convenor and moderator, Year of Basic Sciences for Sustainable Development: Embedding Ethics in Machine Learning, ICTP, Trieste (online)

- ▹ SOC, 'Ethical and Societal Challenges of Machine Learning', ICTP, Trieste (online)
- ▹ SOC, 'Mind the Gap: How We Build, Perceive, Experience and Share Spaces', SISSA
- ▹ SOC, INSAP XI, CalTech

2017– International Executive Committee, 'Inspiration of Astronomical Phenomena' (INSAP)

2018 SOC, Astronomical Data Analysis 9, Valencia

2016 SOC, Astronomical Data Analysis 8, Crete

2015 Co-Chair, 'Identification of Dark Matter with a Cross-Disciplinary Approach' workshop, UAM Madrid

2014 Chair, 'From Cosmology to Customers: Astrostatistics Solutions to Big Data Problems' industry–academia workshop, Imperial

2009, 2013 Chair and founder, 'Cosmostat' conference series: Centro Stefano Franscini, Switzerland (2009)

- ▹ Banff International Research Station (2013)

2013 Lead Organiser, 'Multi-probe Identification of Dark Matter', Kavli ITP programme, UCSB

2012 Chair, DarkAttack12, Centro Stefano Franscini, Switzerland

- ▹ SOC, ICIC inaugural workshop, Imperial College London
- ▹ SOC, 'Statistical Issues in Searches', SLAC, Palo Alto

2010 SOC, RAS National Astronomy Meeting, Glasgow

- ▹ Co-chair, RAS Specialist Discussion Meeting 'Novel Methods for the Exploitation of Large Astronomical and Cosmological Data Sets', London
- ▹ Co-chair, IOP dark matter meeting, London
- ▹ Co-chair, Astrostatistics workshop, London

2012, 2013 Founder and chair, 'The Big Questions' public lecture series (2012) and 'The Sensual Universe' public lecture series (2013), London

2008 Co-chair, 'Bayesian Evidence' workshop, Brighton

2007 Co-chair, RAS Specialist Discussion Meeting 'Statistical Challenges in Particle Astrophysics and Cosmology', London

2002–05 Co-chair, Journées de Cosmologie des Lacs Alpins, Geneva/Annecy/Lausanne/CERN

Teaching

Undergraduate Courses and Lectures

2021 Guest lunchtime lecture, 'The Machine Learning Revolution in Cosmology', Imperial (online)

2015–20 Lecturer, 'Physics of the Universe', 3rd-year physics core course, Imperial

2008–12 Lecturer, 'Statistics of Measurement', 2nd-year physics core course, Imperial

2010–20 Associate lecturer: 'Cosmology', Imperial

2008–09 Associate lecturer: 'Sun, Stars and Planets' (2008–09)

2017, 2018 Guest lecturer, 'Communicating Science', Imperial Horizons course

2016 Guest lecturer, 'Creative Writing', Imperial Horizons course

2010 Guest lecturer, 'Astrophysics', 4th-year option course, Imperial

Postgraduate Courses at SISSA

2023– Bayesian Methods I, Theoretical and Scientific Data Science PhD programme

2020– Ethical Aspects of AI, Theoretical and Scientific Data Science PhD programme

2021– Intelligenza Artificiale e Informazione (with Elisabetta Tola), Master in Science Communication

2020–23 Bayesian Methods II, Theoretical and Scientific Data Science PhD programme

2021– Bayesian Inference and Machine Learning in Cosmology, Theoretical and Scientific Data Science PhD programme

Postgraduate Lectures, Masterclasses and Graduate Schools

2026 AI training course, SISSA; EFC network AI training workshop, Bertinoro

2024– Machine Learning for Business Analytics, MBA in International Business, MIB Trieste

2023 Introduction to Bayesian Inference, SISSA Master in HPC, Trieste; Machine Learning for Business Analytics, MBA in International Business, MIB Trieste

2022	Introduction to Data Science, International Master in Physics of Complex Systems, ICTP, Trieste; Statistics lectures, Cosmology Summer School, ICTP, Trieste
2021	Avventure fra Arte e Scienza nel Comunicare la Cosmologia, SISSA MSC lecture; From Bayes to Machine Learning, Heidelberg Grad Days (online); Bayesian Inference, Saas-Fee Course on 'Astronomy in the Era of Big Data' (online)
2019	Keynote, Research Computing Summer School, Imperial College London
2018	ICIC Data Analysis Postgraduate School, Imperial College London; Analytics, Inference and Computation in Cosmology: Advanced Methods, Cargèse, France; Astronomical Data Analysis IX Summer School, Valencia, Spain
2017	Stats4Astro Astrostatistics School, Autrans, France; ISAPP Summer School, Texel, The Netherlands; First Italian Astrostatistics School, Milan
2016	European Space Astronomy Centre (ESAC) Statistics Workshop, Villanueva de la Cañada, Spain; ADA8 Summer School on Astronomical Data Analysis, Chania, Greece
2014	XII School of Cosmology, Cargèse, France; Niels Bohr Institute PhD School, 'Statistical Methods', Copenhagen; XXVI Winter School, 'Bayesian Inference in Astronomy and Astrophysics', Canary Islands; ICIC STFC Data Analysis School, London; 44th Saas-Fee Advanced Course on Astronomy and Astrophysics, 'Cosmology with Wide-Field Surveys', Switzerland
2013	School on Bayesian Analysis in Physics and Astronomy, Stellenbosch, South Africa; ICIC Data Analysis Postgraduate School, London
2012	'Statistical Inference', African Institute for Mathematical Sciences postgraduate course, Cape Town
2011	'Advanced Statistical Methods for Cosmology and Astroparticle Physics', Niels Bohr Institute, Copenhagen
2010	Cosmology postgraduate lectures, Imperial; 'Advanced Statistical Methods' lectures, Imperial; 'Statistical Methods for Cosmology', 1st Jayme Tiomno School of Cosmology, Rio de Janeiro
2009	'Advanced Statistical Tools for Cosmology', Valencia University; 'Probability Theory, Statistics and Data Analysis' postgraduate lectures, Imperial/UCL; 'The CMB and its Statistical Interpretation', Würzburg University postgraduate lecture series; 'Modern Cosmology' postgraduate lectures, Imperial/UCL; 'Advanced Statistical Tools for Cosmology' postgraduate lectures, U. Autónoma Madrid

Supervision

Postdoctoral Researchers

2025–	Giulio Scelfo
2024–26	Junsong Cang
2024–25	Martin de los Rios (now faculty at CONICET, Argentina)
2024–25	David Prelogović
2023–	Rahul Srinivasan
2023–25	Chiara Moretti
2023	Matteo Breschi
2022–	Andre Scaffidi
2022–24	Gabriella Contardo
2020–22	Ira Wolfson
2020–21	Victor Bonjean
2018–20	Eliel Camargo-Molina (now faculty at U. Uppsala)
2017–20	Alex Geringer-Sameth (now permanent staff at Lawrence Livermore National Lab)
2014	Kaisey Mandel (now faculty at Cambridge)

Current PhD and MSc Students

2023–	M. Rigo, S. Mishra, M. Giunta, U. Zivanovic (MSc)

Completed PhD Students (Thesis Title; Distinctions and First Destination)

2021–25	M. Geng (SISSA), *Reflections on Distributions: Humans, Machines, Languages* (postdoc at École Normale Supérieure, Paris)
2020–24	K. Karchev (SISSA), *SNIa Cosmology for the 21st Century* – IAU Best Thesis Prize: Facilities, Technologies and Data Science 2025; Springer Thesis Prize 2025) (Simons Foundation Fellow, Barcelona)
2020–23	M. Autenrieth (Imperial Statistics, co-supervision), *Principled Bayesian Modelling and Statistical Learning with Non-Representative Data in Astrophysics* – Royal Statistical Society Emerging Applications Thesis Prize 2025 (postdoc at Imperial)
2017–22	W. Rahman (Imperial), *Constraining the Anisotropic Expansion of the Universe with Type Ia Supernovae and Improving the Treatment of Selection Effects within Bayesian Hierarchical Models* (data scientist at Meta)

2015–19	S. Hoof (Imperial, co-supervision), *Global Fits of Axions and WIMPs in Astrophysics, Cosmology and Particle Physics* (Imperial President Scholarship holder; postdoc at Göttingen)
2014–18	J. McKay (Imperial, co-supervision), *Constraining Dark Matter with Renormalisation and Global Fits* (Imperial President Scholarship holder; software developer at Verizon Connect)
2013–17	H. Shariff (Imperial), *Application of Bayesian Statistics in Supernovae Ia Cosmology* (investment consultant at Mercer)
2011–14	C. Strege (Imperial), *Characterisation of Particle Dark Matter via Multiple Probes* — Winton Capital best PhD thesis in physics 2014 (quant at CityBank)
2008–11	M. March (Imperial), *Advanced Statistical Methods for Astrophysical Probes of Cosmology* — Springer Thesis Prize 2011, Winton Capital best PhD thesis in physics 2011 (postdoc at Penn State)

MSc / MSci Students

2025	U. Zivanović (U. Trieste), *An Application of the Rotary Masked Autoencoder to Irregular Sparse Astrophysics Datasets*
2023	R. Serra (U. Trieste), *Improving Photometric Galaxy Redshift Estimation under Covariate Shift with StratLearn*
2022	S. di Gioia (U. Trieste), *Fast and Parallel Algorithms for Causal Discovery in Astrophysics*
2021	R. Corti (U. Trieste), *Machine Learning Methods for Supernova Type Ia Selection Effects Modelling*
2019	M. von Wietersheim-Kramsta (Imperial), *Measuring the Galaxy Mass Function Using Photometric Data*
2017	H. Bouvier (Imperial), *Determining Type Ia Supernovae Host Galaxy Characteristics Using Galaxy Zoo: Hubble and the Prospector-α Model*
2017	S. Kobayashi (Imperial), *Determining Type Ia Supernovae Host Galaxy Characteristics Using Galaxy Zoo: Hubble and the Prospector-α Model*
2016	E. Revsbech (Imperial Statistics, co-supervision), *Synthetically Augmented Light Curve Classification* — Winton Capital Prize Best Project; I. Siska (Imperial), *Constraining Anisotropic Cosmological Expansion using Supernovae Type Ia within a Bayesian Statistical Framework*; A. Monge Imedio (Imperial), *Constraining Cosmological Anisotropies with SNIa Data Using Bayesian Methods*
2015	S. Pak (Imperial), *Constraining Cosmic Expansion Anisotropy using Type Ia Supernovae*; K. Blanchette (Imperial), *Cosmological Parameter Inference in a Bayesian Hierarchical Model with Redshift-Dependent Hubble Residuals*
2011	G. Pisani (co-supervision, Rome La Sapienza)

Other Supervision

2011–20	35 BSc projects, Imperial
	22 Summer interns / UROP / IROP / ICTP / UWC undergraduate students supervised

Mentorship

2026–	Active member, LSST/DESC mentoring programme

Examining at Postgraduate Level

External Examiner / Assessor

2026	A. Bairagi (PhD, Institut d'Astrophysique de Paris)
2025	V. Blasone (PhD, U. Trieste, Applied Data Science and AI programme)
2024	Z. Rokavec (MSc, U. Nova Gorica); A. Prajapati (PhD, Gran Sasso Science Institute); D. Prelogović (PhD, Scuola Normale Superiore; committee member); S. Rinaldi (PhD, U. Pisa)
2023	D. Lanzieri (PhD, CEA Paris-Saclay)
2022	J. Thorp (PhD, Cambridge); F. Agocs (PhD, Cambridge)
2019	R. Hardwick (PhD, U. Portsmouth)
2018	J. Faulkner (PhD, U. Edinburgh)
2017	S. Hee (PhD, U. Cambridge); H. Mootoovaloo (MSc, U. Cape Town)
2016	T. Walwyn (MSc, U. Cape Town)
2014	F. Vansyngel (PhD, Institut d'Astrophysique de Paris)
2013	H. Silverwood (MSc, U. Canterbury, NZ)
2011	Y. Akrami (PhD, Stockholm University; opponent); J. Newling (MSc, U. Cape Town)

Internal Examiner, Imperial College London

2022	E. Hauke (PhD, Centre for Higher Education Research and Scholarship)
2017	Shijing Si (PhD, Statistics Section)
2016	J. Alsing (PhD, Astrophysics group)
2013	H. Shariff (MSc, Astrophysics group)

Invited Scientific Talks

2026 Plenary talk, EUCAIFCon, Heidelberg
- Physics dept, Ljubljana U.
- Data science seminar, CERN

2025 Panelist, 'Agentic AI, LLMs and Research', SISSA, Trieste
- Keynote, IFPU Focus Week, Trieste
- Keynote, Randstad Foundation think tank meeting, Milan
- Keynote, Ital-IA, Italian National meeting on AI, Trieste

2024 Panel, Climate Conference Arena, Bologna
- Colloquium, Celebrating 60 Years of the Associates Programme, ICTP, Trieste
- AI panel moderator, ICTP Symposium 'The Future of Scientific Computing', Trieste
- Scientific Afternoons of the University of Nova Gorica, Nova Gorica
- CMSP News and Views Colloquium, ICTP, Trieste
- Erwin Schrödinger Colloquium, University of Zürich
- Keynote, Research in Dialogue, University of Bern
- Heidelberg Joint Astronomical Colloquium

2023 Astrophysics Colloquium, Ludwig-Maximilians-Universität, Munich
- Colloquium Series in Theoretical and Computational Physics (CSTCP)

2022 'StratLearn: A General-Purpose Statistical Method for Improved Learning under Covariate Shift', IAA/IAU Astroinformatics and Astrostatistics Seminars (online)

2021 'The Impact of Machine Learning in Cosmology' (online)

2019 Plenary, Picturing the Invisible Conference, UAL, London
- Keynote, 'Artificial Intelligence: Art or Science?', SISSA, Trieste
- Keynote, ICTP, Trieste
- AHRC Network meeting

2017 Colloquium, University of Bern
- Colloquium, Indian Institute of Technology Madras

2016 Keynote, Global Challenges Showcase event, Imperial College London
- Keynote, COSMO21 Conference, Chania, Greece

2015 Southampton Physics Colloquium
- BASP Frontiers 2015, Villars, Switzerland

2013 Director's Blackboard Talk, Kavli Institute, Santa Barbara

2012 Invited plenary, Statistical Issues in Searches, SLAC, Stanford
- Invited plenary, Astronomical Data Analysis 7, Cargèse
- Invited plenary, MaxEnt 2012, Garching

2011 Invited plenary, 58th ISI World Congress, Dublin
- Invited plenary, Statistical Challenges in Modern Astronomy V, Penn State
- Invited talk, IOP dark matter meeting, London
- Invited plenary, Aspen Winter Conference, 'Indirect and Direct Detection of Dark Matter', Aspen
- Colloquium, AIMS, Cape Town

2010 Colloquium, EPFL
- Invited plenary, PROSPECT workshop, Stockholm
- Summary talk, 'Statistical Issues Relevant to Significance of Discovery Claims', Banff
- Invited plenary, TeV Particle Astrophysics, Paris

2009 Review talk, ILIAS/ENTApP meeting, CERN, Geneva
- Invited talk, National Astronomy Meeting, Hatfield

2008 Michelson Prize Lecture, Case Western Reserve University, USA
- Invited plenary, ESF Exploratory Workshop, Oporto
- Invited talk, Tools08, Munich
- Review talk, XXVI Workshop on Recent Developments in High Energy Physics and Cosmology, Ancient Olympia

2007 Lord Kelvin Award Lecture
- Invited talk, TeV Particle Astrophysics 2007, Venice

2006 Invited talk, 11th Marcel Grossmann Meeting, Berlin
- Invited plenary, RAS General Meeting, London

2005 Review talk, ‘Recent Advances in Particle Physics and Cosmology’, Thessaloniki

Contributed Conference Talks

2024 The Dark Matters, parallel session at the RGS-IBG Annual International Conference (online)
2021 Debating the potential of machine learning in astronomical surveys (online)
2019 CosmoAI, Ascona, Switzerland
2018 Cosmo21, Valencia
2017 The Road to the Stars, Santiago de Compostela
2016 XENON100 Collaboration Meeting, Mainz, Germany
2013 ISI World Statistics Meeting, Hong Kong
2012 ISBA World Meeting, Kyoto
2011 ERC Kickoff meeting, Amsterdam
- IOP Dark Matter Meeting at King’s College
- 58th ISI World Congress
- PHYSTAT2011, CERN, Geneva

2009 Corfu Summer Institute, Greece
- cosmostats09, Ascona, Switzerland
- Galileo Galilei Institute dark energy conference, Florence

2008 Nested Sampling workshop, Cambridge
- Bayesian evidence workshop, University of Sussex
- CosmoTools08, Marseille

2007 Initial Conditions in Cosmology workshop, Würzburg, Germany
- CERN dark matter workshop, Geneva

2006 RAS Young Astronomers Meeting, London
- Identification of Dark Matter Conference, Rhodes, Greece
- Galileo Galilei Institute Cosmology Workshop, Florence
- The Dark Side of the Universe Workshop, Madrid
- CosmoUK Meeting, Oxford
- Francesco Melchiorri Memorial Conference, Rome
- TeV Particle Astrophysics II, Madison, Wisconsin
- Astroparticle UK Meeting, Sheffield
- National Astronomy Meeting, Leicester
- SKA conference, Oxford
- St Anne’s College, Oxford

2005 European Dark Energy Network Meeting, Paris
- Subaru/Gemini dark energy meeting, Waikoloa, Hawaii
- PHYSTAT05, Oxford
- Sils Maria Cosmology Workshop, Engadin
- ENTApP meeting, CERN, Geneva

2004 7th Journée des Lacs Alpins de Cosmologie, EPFL Lausanne
- 6th Journée de Cosmologie des Lacs Alpins, Geneva

2003 Second Aegean Summer School on the Early Universe, Syros
- Sils Maria Cosmology Workshop, Engadin
- ATLAS Collaboration Physics Tutorial, CERN
- X Marcel Grossmann Meeting, Rio de Janeiro
- CMBNet Workshop on Science and Parameter Extraction, Oxford

2002 5th Journée de Cosmologie des Lacs Alpins, Annecy
- CMBNet Workshop, Geneva
- Santa Fe Cosmology Workshop, New Mexico

2001 CMBNet General Meeting, Frascati, Italy
- UK Cosmology Meeting, Ambleside

Competitive Grants — Science & Knowledge Transfer

Overall total: *£10.4M • €2M*
Grants above 100k (in either currency) are highlighted in bold.

1. *Optimal inference from radio images of the epoch of reionization, Co-I* (2022). **€192k**
   Awarding body: Ministero dell'Università e della Ricerca, Prin programme; PI: A. Mesinger.
2. *Centro Nazionale Future Artificial Intelligence Research (FAIR), Spoke Edge and Exascale computing, Co-I (1 postdoc & salary co-fund)* (2022). **€520k**
   Awarding body: PNRR, Italian Government; SISSA lead: G. Sanguinetti.
3. *Centro Nazionale HPC and Quantum Computing, Spoke COSMOS, Co-I (1 postdoc & salary co-fund)* (2022). **€565k**
   Awarding body: PNRR, Italian Government; SISSA lead: M. Viel.
4. *Data Science methods for Multi-Messenger Astrophysics & Multi-Survey Cosmology, PI (1 postdoc)* (2022). **€140k**
   Awarding body: PRO3 Programma congiunto.
5. *Particle Physics Theory Consolidated Grant, co-I (1 postdoc & salary co-fund)* (2016). **£6.2M**
   Awarding body: STFC; PI: J. Gauntlett.
6. *Astrophysics Consolidated Grant, co-I (1 postdoc & salary co-fund)* (2016). **£2.4M**
   Awarding body: STFC; PI: S. Warren.
7. *Impact Acceleration Grant, PI* (2016). £46k
   Awarding body: EPSRC.
8. *Impact Acceleration Award, PI* (2014). £15k
   Awarding body: STFC.
9. *Pathways to Impact funding, PI* (2013). £14k
   Awarding body: EPSRC.
10. *Astrophysics Consolidated Grant, Co-I (1 postdoc & salary co-fund)* (2012). **£463k**
    Awarding body: STFC; PI: A. Jaffe.
11. *Sir Norman Lockyer Fellowship, PI* (2005). **£150k**
    Awarding body: Royal Astronomical Society.
12. *Postdoctoral Fellowship, PI (declined)* (2005). .... £ 90k
    Awarding body: European Space Agency.

Subtotal: *£9.29M • €1.42M*

Competitive Grants — Science Communication & Education

13. *Erasmus Mundus programme development support, co-I* (2026). €60k
    Awarding body: EU.
14. *Incentivi Triennali alle Iniziative di Divulgazione, Festival Scienza & Virgola, PI* (2026). **€200k**
    Awarding body: Regione FVG.
15. *Investigative journalism grant, co-I* (2025). €50k
    Awarding body: Journalism Science Alliance.
16. *Bando Divulgazione Cultura Scientifica, PI* (2025). €30k
    Awarding body: Regione FVG.
17. *Stelle Scadenti: theatre play production, PI* (2021). €25k
    Awarding body: Regione Autonoma Friuli Venezia Giulia.
18. *The Edge of the Sky storytelling project, co-I* (2020). £12k
    Awarding body: Creative Scotland.
19. *Pedagogy Transformation Grant, co-I* (2018). **£451k**
    Awarding body: Education Office, Imperial College London.
20. *Public Engagement Grant, PI* (2016). £1.5k
    Awarding body: Royal Astronomical Society.
21. *Public Engagement Grant, PI* (2016). £2k
    Awarding body: Institute of Physics.
22. *Public Engagement Fellowship, PI (salary co-fund)* (2013). **£106k**
    Awarding body: STFC.
23. *Public Engagement Grant, PI* (2011). £8.5k
    Awarding body: STFC.

Subtotal: *£581k • €365k*

Travel Grants, Student Fellowships and Conference Support Grants

24. *Visiting Scientist Travel Grant, PI* (2026). €4k
Awarding body: SISSA.
25. *Visiting Scientist Travel Grant, PI* (2024). €10k
Awarding body: SISSA.
26. *PhD studentship Dottorati Green/Innovazione/DM 351, PI* (2023). **€100k**
Awarding body: PNRR, Italian Government.
27. *PhD studentship PON Innovazione, PI* (2021). **€100k**
Awarding body: PNRR, Italian Government.
28. *PhD studentship, PI* (2017). £70k
Awarding body: STFC.
29. *Development of Novel Statistical Tools for the Analysis of Astronomical Data, Marie Skodowska-Curie Research and Innovation Staff Exchange, co-I* (2016). **£107k**
Awarding body: European Commission.
30. *Travel and Exchange Grant, PI* (2016). £12k
Awarding body: Royal Society.
31. *SPRINT Collaboration Grant, PI* (2016). £10k
Awarding body: FAPESP-Imperial.
32. *Conference Organization Grant, PI* (2015). £35k
Awarding body: IFT, Madrid.
33. *Data Analysis Summer School funding, Co-I* (2014). £23k
Awarding body: STFC.
34. *Search for Dark Matter particles with the XENON experiment collaboration exchange, PI* (2013). £6k
Awarding body: Imperial College Trust.
35. *PhD studentship, PI* (2013). £70k
Awarding body: STFC.
36. *Programme Organization Grant, PI* (2013). £65k
Awarding body: KITP, Santa Barbara.
37. *Travel and Exchange Grant, PI* (2012). £12k
Awarding body: Royal Society.
38. *CosmoClassic Cross-Disciplinary Collaboration, Co-I* (2012). £16k
Awarding body: Faculty Strategic Research Fund, Imperial College London.
39. *Conference Organization Grant, Co-I* (2012). £25k
Awarding body: Centre Stefano Franscini, ETHZ.
40. *Summer Student Project Bursaries, PI* (2011). £3k
Awarding body: UROP.
41. *PhD Studentship Extension Funding, PI* (2011). £3k
Awarding body: Royal Astronomical Society.
42. *Travel Grant, PI* (2010). £2k
Awarding body: Royal Astronomical Society.
43. *PhD studentship, PI* (2008). £70k
Awarding body: STFC.

Subtotal: *£529k • €214k*

Scientific Publications

ORCID: 0000-0002-3415-0707 | Scopus: 7004543998 | Web of Science: AAN-7820-2020 | iNSPIRE: R.Trotta.1 | Google Scholar: V5FvaSYAAAAJ | NASA ads

115 published papers in peer-reviewed physics journals or machine learning conferences (12 of which as first author, including a single-author review article and 3 in *Phys. Rev. Lett.*); 18 conference proceedings; 5 white papers and editorial works; 5 chapters in academic books.

Citation count on August 7, 2026: 14,500+, h-index: 54 (INSPIRE/NASA ADS, higher of each per paper); 13,000+, h-index 57 (Google Scholar).

indicates a paper with more than 100 citations (22). 100+
indicates a paper with more than 500 citations (2). 500+
indicates a paper with more than 1000 citations (3). 1000+
Co-authors who at the time of publication were my PhD or Master students are underlined.

Refereed Physics Journals

1. J. Aalbers et al (DARWIN Collaboration; A. Scaffidi & **R. Trotta** as corresponding authors, 2026), 'Model-independent searches of new physics in DARWIN with a semi-supervised deep learning pipeline', *Europ. Phys. J. C, 86, 312 (2026)*Eur. Phys. J. C 86, 312 (2026), `arxiv:2410.00755`.
2. S. Mishra, **R. Trotta** & M. Viel (2026), 'Cosmo-FOLD: Fast generation and upscaling of field-level cosmological maps with overlap latent diffusion', submitted, `arxiv:2601.14377`
3. M. de los Rios, S. di Gioia, F. Iocco & **R. Trotta** (2025), 'Dark Matter profiles of "in silico" galaxies: deep learning inference', submitted, `arxiv:2510.18964`
4. K. Karchev, **R. Trotta** & R. Jimenez (2026), 'CIGaRS I: Combined simulation-based inference from SNae Ia and host photometry[1]', *Nature Astronomy* in print, `arxiv:2508.15899`
5. R. Srinivasan, E. Barausse, N. Korsakova & **R. Trotta** (2025), 'Simulation-based population inference of LISA's Galactic binaries: Bypassing the global fit', *Phys. Rev. D* 112, 103043 (2025), `arxiv:2506.22543`
6. S. Mishra, **R. Trotta** & M. Viel (2026a), 'Large, fast and accurate HI intensity maps with latent overlap diffusion', *Mon. Not. R. Astron. Soc.* , 545, 3 (2026), `arxiv:2506.08086`
7. C. Moretti, M. Autenrieth, R. Serra, **R. Trotta**, D.A. van Dyk & A. Mesinger (2025), 'StratLearn-z: Improved photo-z estimation from spectroscopic data subject to selection effects', *The Open Journal of Astrophysics 8 (May)*, `arxiv:2409.20379`
8. S. Springer, A. Scaffidi, M. Autenrieth, G. Contardo, A. Laio, **R. Trotta** & H. Haario (2025), 'Detecting Localized Density Anomalies in Multivariate Data via Coin-Flip Statistics', submitted, `arxiv:2503.23927`
9. G. Contardo, **R. Trotta**, S. di Gioia, D.W. Hogg & F. Villaescusa-Navarro (2025), 'On the effects of parameters on galaxy properties in CAMELS and the predictability of Ω_m', *ApJ* 88 (2025) 1, 54, `arxiv:2503.22654`
10. A. Lüber, K. Karchev, C. Fisher, M. Heim, **R. Trotta** & K. Heng (2025), 'Near-instantaneous Atmospheric Retrievals and Model Comparison with FASTER', *ApJL* 984 L32, `arxiv:2502.18045`
11. M. Rigo, **R. Trotta** & M. Viel (2025), 'JERALD: high-fidelity dark matter, stellar mass and neutral hydrogen maps from fast N-body simulations', *Mon. Not. R. Astron. Soc.* 541,1, 166–178 (2025), `arxiv:2501.09168`
12. J. Cang, A. Mesinger, S.G. Murray, D. Breitman, Y. Qin & **R. Trotta** (2025), 'The EDGES measurement disfavors an excess radio background during the cosmic dawn', *A& A*, 698, A152 (2025), `arxiv:2411.08134`
13. Aalbers et al (The XLZD Collaboration, incl. **R. Trotta**, 2025b), 'The XLZD Design Book: Towards the Next-Generation Liquid Xenon Observatory for Dark Matter and Neutrino Physics', *Eur. Phys. J.* C (2025) 85: 1192, `arxiv:2410.17137` (Role: contributing author)
14. J. Aalbers et al (The XLZD Collaboration, incl. **R. Trotta**, 2025a), 'Neutrinoless Double Beta Decay Sensitivity of the XLZD Rare Event Observatory', *J. Phys. G: Nucl. Part. Phys.* 52 045102, `arxiv:2410.19016` (Role: contributing author)
15. J. Aalbers et al (The XLZD Collaboration, incl. **R. Trotta**, 2024), 'The XLZD Design Book: Towards the Next-Generation Liquid Xenon Observatory for Dark Matter and Neutrino Physics', submitted to Eur. Phys. J. C., `arxiv:2410.17137`. (Role: contributing author)
16. K. Karchev & **R. Trotta** (2024), 'STAR NRE: Solving supernova selection effects with set-based truncated auto-regressive neural ratio estimation', *JCAP* in print, `arxiv:2409.03837`
17. M. Benito, K. Karchev, R.K. Leane, S. Poder, J. Smirnov & **R. Trotta** (2024), 'Dark Matter Halo Parameters from Overheated Exoplanets via Bayesian Hierarchical Inference', *JCAP* 07(2024)038, `arxiv:2405.09578`
18. K. Karchev, M. Grayling, B. Boyd, K. Mandel, **R. Trotta** & C. Weniger (2024), 'SIDE-real: Truncated marginal neural ratio estimation for Supernova Ia Dust Extinction with real data', *Mon. Not. R. Astron. Soc.* , 530, 4, 3881–3896 (2024), `arxiv:2403.07871`

[1] Finalist at the JSM 2026 Astrostatistics Interest Group Student Paper Competition

19. M. Adrover et al (The DARWIN Collaboration, incl. **R. Trotta**, 2024), 'Cosmogenic background simulations for the DARWIN observatory at different underground locations', *Eur. Phys. J. C* 84, 88 (2024), `arxiv:2306.16340`. (Role: contributing author)

20. R. Srinivasan, M. Crisostomi, **R. Trotta**, E. Barausse & M. Breschi (2024), 'Bayesian Evidence estimation from posterior samples with normalizing flows', *Phys. Rev. D* , 110.123007(2024), `arxiv:2404.12294`

21. S. Maitra, S. Cristiani, M. Viel, **R. Trotta** & G. Cupani (2024), 'Parameter estimation from Lyα forest in Fourier space using Information Maximising Neural Network', *A&A*, 690, A154 (2024), `arxiv:2404.04327`

22. M. Autenrieth, A. H. Wright, **R. Trotta**, D.A. van Dyk, D.C. Stenning & B. Joachimi (2024), 'Improved Weak Lensing Photometric Redshift Calibration via StratLearn and Hierarchical Modeling[2]', *Mon. Not. R. Astron. Soc.* 534, 4, 3808–3831, `arxiv:2401.04687`.

23. D. Breitman, A. Mesinger, S. Murray, D. Prelogovic, Y. Qin & **R. Trotta** (2024), '21cmEMU: an emulator of 21cmFAST summary observables', *Mon. Not. R. Astron. Soc.* , 527, 4, 9833–9852, `arxiv:2309.05697`

24. M. Autenrieth, D. van Dyk, **R. Trotta** & D. Stenning (2023), 'Stratified Learning: A General-Purpose Statistical Method for Improved Learning under Covariate Shift[3]', *Anal. Data Min.: ASA Data Sci. J.* 1-16, `arxiv:2106.11211`.

25. M. Crisostomi, K. Dey, E. Barausse & **R. Trotta** (2023), 'Neural Posterior Estimation with guaranteed exact coverage: the ringdown of GW150914', *Phys. Rev. D* , 108, 044029 (2023), `arxiv:2305.18528`

26. A. Heavens, A. Mootoovaloo, **R. Trotta** & E. Sellentin (2023), 'Extreme data compression for Bayesian model comparison', *JCAP* 11(2023)048, `arxiv:2306.15998`

27. K. Karchev, **R. Trotta** & C. Weniger (2023), 'SICRET: Supernova Ia Cosmology with truncated marginal neural Ratio EsTimation', *Mon. Not. R. Astron. Soc.* , 520 (2023) 1056-1072, `arxiv:2209.06733`

28. J Aalbers et al (DARWIN Collaboration, incl. **R. Trotta**, 2023), 'A next-generation liquid xenon observatory for dark matter and neutrino physics', *J. Phys. G: Nucl. Part. Phys. 50* (2023) 013001, 10.1088/1361-6471/ac841a. (Role: contributing author) 100+

29. L. Althueser et al. (The DARWIN Collaboration, incl. **R. Trotta**, 2022), 'GPU-based optical simulation of the DARWIN detector', *Journal of Instrumentation*, 17, 07 P07018, 10.1088/1748-0221/17/07/P07018. (Role: contributing author)

30. S.S. AbdusSalam, F.J. Agocs, B.C. Allanach, P. Athron, C. Balázs, E. Bagnaschi, P. Bechtle, O. Buchmueller, A. Beniwal, J. Bhom, S. Bloor, T. Bringmann, A. Buckley, A. Butter, J. Eliel Camargo-Molina, M. Chrzaszcz, J. Conrad, J.M. Cornell, M. Danninger, J. de Blas, A. De Roeck, K. Desch, M. Dolan, H. Dreiner, O. Eberhardt, J. Ellis, B. Farmer, M. Fedele, H. Flächer, A. Fowlie, T.E. Gonzalo, P. Grace, M. Hamer, W. Handley, J. Harz, S. Heinemeyer, S. Hoof, S. Hotinli, P. Jackson, F. Kahlhoefer, K. Kowalska, M. Krämer, A. Kvellestad, M. Lucio Martinez, F. Mahmoudi, D. Martinez Santos, G.D. Martinez, S. Mishima, K. Olive, A. Paul, M.T. Prim, W. Porod, A. Raklev, J.J. Renk, C. Rogan, L. Roszkowski, R. Ruiz de Austri, K. Sakurai, A. Scaffidi, P. Scott, E.M. Sessolo, T. Stefaniak, P. Stöcker, W. Su, S. Trojanowski, **R. Trotta**, Y.-L. Sming Tsai, J. Van den Abeele, M. Valli, A.C. Vincent, G. Weiglein, M. White, P. Wienemann, L. Wu, Y. Zhang (2022), 'Simple and statistically sound strategies for analysing physical theories', *Rept. Prog. Phys.* 85 (2022) 052201, 10.1088/1361-6633/ac60ac. (Role: contributing author)

31. W. Rahman, **R. Trotta**, S. S. Boruah, M. J. Hudson & D. A. van Dyk (2022), 'New Constraints on Anisotropic Expansion from Supernovae Type Ia', *Mon. Not. R. Astron. Soc.* 514, 1 (2022), 139–163, doi.org/10.1093/mnras/stac1223.

32. J. Aalbers, et al (The DARWIN Collaboration, incl. **R. Trotta**, 2020b), 'Solar Neutrino Detection Sensitivity in DARWIN via Electron Scattering', *Eur. Phys. J. C* (2020) 80: 1133, `arxiv:2006.03114`. (Role: contributing author)

33. S. Ando, A. Geringer-Sameth, N. Hiroshima, S. Hoof, **R. Trotta** & M.G. Walker (2020), 'Structure Formation Models Weaken Limits on WIMP Dark Matter from Dwarf Spheroidal Galaxies', *Phys. Rev. D* 102, 061302(R), `arxiv:2002.11956`

34. F. Agostini, (DARWIN Collaboration, incl. **R. Trotta**, 2020a), 'Sensitivity of the DARWIN observatory to the neutrinoless double beta decay of ^{136}Xe', Eur. Phys. J. C 80, 808 (2020), `arxiv:2003.13407`. (Role: contributing author)

35. E.V. Karukes, M. Benito, F. Iocco, **R. Trotta** & A. Geringer-Sameth (2020), 'A robust estimate of the Milky Way mass from rotation curve data', *JCAP* 05(2020)033, `arxiv:1912.04296`

36. S. Hoof, A. Geringer-Sameth & **R. Trotta** (2020), 'A Global Analysis of Dark Matter Signals from 27 Dwarf Spheroidal Galaxies using 11 Years of Fermi-LAT Observations', *JCAP* 2, 012 (2020), `arxiv:1812.06986` 100+

37. E.V. Karukes, M. Benito, F. Iocco, **R. Trotta** & A. Geringer-Sameth (2019), 'Bayesian reconstruction of the Milky Way dark matter distribution', *JCAP* 09, 046 (2019), `arxiv:1901.02463`

38. L. Amendola et al. (Euclid Theory Working Group, incl. **R. Trotta**, 2018), 'Cosmology and Fundamental Physics with the Euclid Satellite', *Living Rev. Relativ.* 21, 2 (2018) `arxiv:1606.00180`. (Role: contributing author, astrostatistics section lead)

39. R. Hill, H. Shariff, **R. Trotta** *et al.* (2018), 'Projected distances to host galaxy reduce SNIa dispersion', *Mon. Not. R. Astron. Soc.* 481, 2 (2018), `arxiv:1612.04417`

40. H.A. Clark, P. Scott, **R. Trotta** & G.F. Lewis (2018), 'Dark matter substructure cannot explain properties of the Fermi Galactic Centre excess', *JCAP* 07, 060 (2018), `arxiv:1612.01539`

[2]Winner of the JSM 2024 Astrostatistics Interest Group Student Paper Competition
[3]Finalist at the JSM 2022 Astrostatistics Interest Group Student Paper Competition

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18. **R. Trotta** (2003), 'The cosmological constant and the paradigm of adiabaticity', *New Astron. Rev.*, 47, 769–774, `arxiv:astro-ph/0304525`

Book Reviews

1. **R. Trotta** (2014), 'Practical Statistics for Astronomers, 2nd Ed (Wall and Jenkins, CUP, 2012)', AIAA Journal, 52, 9, (2014), 2098, doi.org/10.2514/1.J053573
2. **R. Trotta** (2012), 'Modern Statistical Methods for Astronomy (Feigelson and Babu, CUP 2012)', Mathematical Reviews, MR2963292
3. **R. Trotta** (2008), 'Statistical challenges in modern astronomy IV (ASP Conference Series, Vol 371)', The observatory, May 2008
4. **R. Trotta** (2008), 'The Oxford book of modern science writing (Richard Dawkins, Ed, OUP 2008)', Oxford Today, 21, 1 (2008), 53

KNOWLEDGE TRANSFER

Board Memberships and Advisory Roles

2026–	Board of Directors, Fondazione Internazionale Trieste per il Progresso e la Libertà delle Scienze
2025–	Scientific Advisory Board, FACTA.eu
2024–	Scientific Advisory Board, RACHAEL S.r.l., `rachael.swg.it`
2021–24	Chairman, Scientific Advisory Board, RACHAEL S.r.l., `rachael.swg.it`
2018–20	Non-executive Director, Imperial Consultants, `imperial-consultants.co.uk`
2012–20	Director and co-founder, Data Fusion Consultants Ltd
2010–	Consultant, Imperial Consultants

Areas of Expertise

Expertise. Statistical inference and model selection. Data science and machine learning. Numerical data analysis and data mining. Environmental assessment of risk. Astrophysical phenomena, theoretical physics, space travel and cosmology.

Services. Data analysis and interpretation. Probabilistic modeling. Inference, prediction and optimization. Training and workshops: probability theory, inference, numerical methods, computational solutions, data science, machine learning. Expert witness. Scientific consulting for museums and creative arts industry.

Clients

Scientific Consultancy. Beluga Group Ltd (2025); Savio Macchine Tessili (2024); Museo del Castello di Miramare, exhibit *Kosmos* (2023); Oxford University Press (2023, credited); Red Planet Pictures, *Death in Paradise* (2022, credited); Altin Homes (2021); Regione Autonoma FVG (2021); Nature Publishing Group (2020); Fantastic Productions, Netflix feature film (2019); Imperial College outreach and Royal Albert Hall (2018); Enchanted Lions Books (2016, 2018, 2019, credited); The Polish Cultural Institute, London (2018); Left Bank Pictures, *Origin* (2018, credited); Institute of Physics (2017); Ore Design Jewellery (2015); BBC *Sky at Night* (2015); UP Projects *The Floating Cinema* (2015); The Times Cheltenham Science Festival (2015); Objective Productions *Breaking Magic 1 & 2* (2014); Cinema.ai (2014); Magnum Media *Man vs Expert* (2014); INFRA (2013); Catsnake Productions *The Theory of Everything* (2012, Audience Award, 29th International Short Film Festival Berlin 2013); London Science Museum (2008); Deutsches Museum Munich (2006).

Statistical Consultancy. Clients in the shipping industry, energy industry, property developers, designers, banking and finance, environmental groups, fashion, online commerce.

Keynotes, Lectures and Training

2026	Keynote, Generali Security Council; PKB Academy (Lugano)
2025	Keynote, FINECO Advisory Seminar
2024	Online seminar, Intesa San Paolo
2022	Lecturer, CINECA Academy, up-skilling programme for businesses, Trieste; Programme committee member and lecturer, Generali Business Translators training course, online
2021	Knowledge transfer webinar: the case of *RACHAEL*, a big data startup (online)
2019–21	Imperial College Business School Executive Education: Sberbank Digital Technology Programme (2020, 2021); Naspers Executives Programme (2019)
2018	Imperial Space Lab STFC Showcase Industry Event
2017	Keynote, Data Analysts User Group conference, London
2016	Keynote, iHub opening ceremony, Imperial College London

Public Engagement and Science Communication

Books for the General Public

1. **R. Trotta** (2023), 'Starborn. How the Stars Made Us (And Who We Would Be Without Them)', Basic Books.
Translations: Italian (Il Saggiatore, 2025); Korean (Mirae N Co., 2025); Spanish (Pasado & Presente, 2025); Chinese (simplified characters, Unread Sky, 2026).
Awards and distinctions: BBC Radio 4 'Book of the Week'; BBC Sky at Night 5-star review; 'The 22 best non-fiction and popular science books of 2023' (New Scientist); 'Ten Best Science Books of 2023' (The Smithsonian Magazine); 'Ten Best Books of December 2023' (Christian Science Monitor); 'Fall STEM Reads of 2023' (American Scientist); 2025 Nautilus Book Awards Gold Winner.
2. **R. Trotta** (2014), 'The Edge of the Sky: All You Need to Know about the All-There-Is', Basic Books. The book explains cosmology using only the most common 1,000 words in English. Translations: Korean (Kyobo Centre Books, 2015), German (C.H. Beck, 2015), Catalan (El Magnanim, 2020).
Awards and distinctions: Global Thinker 2014 for 'junking astronomy jargon' (*Foreign Policy*, Nov/Dec 2014); 'Best Science Books of 2014', brainpickings.org; 'Great Popular Science Books 2014', *The Huffington Post*; 'Favorite Physics Books of 2014', *Scientific American*; 'Recommended' by Scientific American (Aug 2014); 'Most anticipated books of 2014', *Publishers Weekly*, July 2014; starred review, *Publishers Weekly*, July 2014.

Book Chapters and Fiction

1. **R. Trotta** (2022c), 'Wunderlich Park', *Tamarind Literary Magazine*, Issue 4, Dec 2022.
2. **R. Trotta** (2022b), 'The Dictionary of Invisible Meanings', In *Picturing the Invisible*, R. Morgan and P. Coldwell (eds), UCL Press (2022)
3. **R. Trotta** (2022a), 'The Invisible Universe', In *Picturing the Invisible*, R. Morgan and P. Coldwell (eds), UCL Press (2022)

4. **R. Trotta** (2021), 'Masters and Servants: The Need for Humanities in an AI-dominated Future', in *The Love Makers*, A. Campbell (ed), Goldsmith Press/MIT Press

Other publications for the General Public

1. **R. Trotta** (2025b), 'Un Mahler di silicio', Trieste a Teatro, Aprile-Maggio 2025, 40–41.
2. **R. Trotta** (2025a), 'The stars have always spoken to us. It's time to listen again', Big Issue Magazine, Jan 13th 2025.
3. **R. Trotta** (2024b), 'Da Asimov a Musk, il nostro destino è fra le stelle?', TuttoScienze, La Stampa, Oct 30th 2024.
4. **R. Trotta** (2024a), 'In search of lost deep time', Tangible Territory, Issue 7, Sept 2024.
5. A. Contessa, P. Del Negro, S. Fantoni, R. Rui & **R. Trotta** (2024), 'Un mare di dati/An ocean of data', in: *Kosmos — Il veliero della conoscenza/The sailing ship of knowledge*, exhibition catalogue, SilvanaEditoriale (2024), 141–149
6. **R. Trotta** (2023), 'What We Lose When We Can't Stargaze', *TIME*, Dec 15th 2023
7. **R. Trotta** (2022b), 'Verso un futuro dominato dall'intelligenza artificiale', *Italian Tech*, Nov 2022
8. **R. Trotta** (2022a), 'Il peso del cielo', Meridiana 276, Mar-Apr 2022, 18–25.
9. **R. Trotta** (2020c), 'Vintage 2020', Tangible Territory, Issue 1, Oct 2020.
10. **R. Trotta** (2020b), 'Diary of an astrophysicist', in *The Covid Diaries: Views from Imperial College*, S. Webster (Ed), online e-zine.
11. **R. Trotta** (2020a), 'From the Big Bang to AI', In *Uncertain Ruins*, Petrel, London, (2020)
12. **R. Trotta** (2019b), 'The Smell of Dark Matter: Dane Mitchell's *Perfume Plumes*', Art in Print, Vol 9, 3, Sept-Oct (2019)
13. **R. Trotta** (2019a), 'The Harmony of the Cosmos', Space Spectacular programme brochure Royal Albert Hall, June 2019
14. **R. Trotta** (2018), 'The Harmony of the Cosmos', Space Spectacular programme brochure Royal Albert Hall, May 2018
15. **R. Trotta** (2015f), 'The Physics of Star Trek', Star Trek: The Ultimate Voyage programme brochure Royal Albert Hall, Nov 1st 2015
16. **R. Trotta** (2015e), 'In so many words: Minute world with big stories to tell', NewScientist online, July 30th 2015
17. **R. Trotta** (2015d), 'In so many words: Don't kill small-life before Red World trip', NewScientist online, July 24th 2015
18. **R. Trotta** (2015c), 'Hoped-for dark matter flash might instead be the corpses of stars', NewScientist online, July 1st 2015
19. **R. Trotta** (2015b), 'In so many words: How to ride the space-wind to the stars', NewScientist online, June 4th 2015
20. **R. Trotta** (2015a), 'Simple is Hard', Army ALT, Jan-Mar 2015
21. **R. Trotta** (2014d), 'Is the All-There-Is all there is?', Significance magazine, December 2014
22. **R. Trotta** (2014c), 'The science of Interstellar: Astrophysics, but not as we know it', The Guardian, Nov 5th 2014
23. **R. Trotta** (2014b), 'Can the universe be described in simple English?', British Council's VOICES invited blog post, Nov 12th 2014
24. **R. Trotta** (2014a), 'Mangalyaan: The Power of Science and Simplicity', The Eastern Eye, Oct 3rd 2014
25. **R. Trotta** (2013), 'Only one Universe to observe', SCOPE, May 2013
26. **R. Trotta** (2012), 'Cosmic Dialogue', People and Science, March 2012 (featured article)
27. **R. Trotta** (2007e), 'The trouble with strings. Interview with Lee Smolin', Newton, November 2007 (cover story)
28. **R. Trotta** (2007d), 'Dark Matter: Probing the Arche-Fossil', COLLAPSE Vol II, March 2007
29. **R. Trotta** (2007c), 'Before the Big Bang', Newton, September 2007 (cover story)
30. **R. Trotta** (2007b), 'Mapping out the invisible Universe', Newton, April 2007 (cover story)
31. **R. Trotta** (2007a), 'The anthropic principle: Interview with John Barrow', Newton, October 2006 (cover story)
32. **R. Trotta** (2006g), 'Cosmologists scoop Nobel Prize for Physics 2006', LeScienze, Nov 2006
33. **R. Trotta** (2006f), 'Giant telescopes of the future: the Square Kilometer Array', LeScienze, July 2006
34. **R. Trotta** (2006e), 'Latest news from the baby Universe', LeScienze, June 2006
35. **R. Trotta** (2006d), 'Interview with James Lovelock', LeScienze, April 2006
36. **R. Trotta** (2006c), 'Listening to cosmic sound', LeScienze, January 2006.
37. **R. Trotta** (2006b), 'Modified gravity', Newton, April 2006
38. **R. Trotta** (2006a), 'Killer asteroids', Newton, January 2006
39. **R. Trotta** (2005c), 'Looking at the Universe through neutrinos', LeScienze, December 2005
40. **R. Trotta** (2005b), 'The mysterious past of our Universe', Newton, October 2005

41. **R. Trotta** (2005a), 'Einstein's legacy', Newton, June 2005

Public Lectures, Festivals, Educational Activities and Talks in Schools

2026 Starborn — CERN Library talk

- Keynote, AstroCalina fundraising event
- Panelist, World Pignolo Day, Udine
- Keynote, AI-peritivo, Trieste

2025 Locarno Palacinema

- Farra d'Isonzo Astronomy Club
- Genova Science Festival
- Festival della Scienza dell'Alto Vicentino, Schio
- PordenoneLegge Book Festival
- Strange Days, GO25!, Gorizia
- Lavagne, Festival della Letteratura, Mantova
- Festival della Letteratura, Mantova
- Salone del Libro di Torino
- 'Cresciuti dalle stelle', conferenza-spettacolo, Festival del libro scientifico Scienza & Virgola, Trieste
- 'Scrivere di scienza', panelist, Festival del libro scientifico Scienza & Virgola, Trieste
- 'Dall'utopia alla tecnocrazia distopica: il ruolo dell'Intelligenza Artificiale nell'ideologia del Muskismo', AI Talks, Trieste
- 'How The Stars Built Our World', Royal Institution, London
- Oxford Literary Festival, Oxford
- Delaware Astronomy Club, online
- Bookpride, Milano
- 'The life of stars', elementary school Il Castelletto, Trieste
- 'Ancient humans and the cosmos', The Prehistory Guys YouTube channel

2024 'Da Isaac ad Elon', MondoFuturo, Trieste

- 'AI e Astrofisica', TriesteNext panelist, Trieste
- MediaLink festival panelist, Trieste
- Star walk, ALSO 2024 Festival
- 'Starborn', ALSO 2024 Festival
- 'AI for Science' public debate moderator, University of Amsterdam
- Festival della Scienza dei Dati e dell'Intelligenza Artificiale (Monfalcone)
- Festival della Scienza e Filosofia (Foligno)
- Scuola Media Galilei, Sant'Eraclio
- Grand Incontri, Miramare Castle (Trieste)
- AI from the Sky, United World College (Duino)

2023 Q&A with Louise Allcock and Robin Ince, RATIO Forum (Sofia)

- Starborn — How the Stars Made Us, RATIO Forum (Sofia)
- Starborn — book launch, in conversation with Claudia de Rham, Imperial College London (London)
- Starborn — book launch, in conversation with Roger Davies, St Anne's College (Oxford)
- Starborn — book launch, in conversation with Carissa Vêliz, Hertford College (Oxford)
- From Babbage to Musk: Promethean Visions of AI, keynote, MIB Trieste School of Management alumni reunion (Trieste)
- Dal Big Bang all'Intelligenza Artificiale, QuarantaScienza, Accademia dei XL (online)
- ChatGPT, do you love me? Panel chair at TriesteNext
- I promemoria di Italo Calvino per affrontare il terzo millennio, discussion panel, Scienza e Virgola festival (Trieste)
- Dal Big Bang all'Intelligenza Artificiale, AITalks series (Trieste)
- Dal Big Bang all'intelligenza artificiale, I venerdì dell'universo (Ferrara)
- Il sistema solare, Scuola Europea "Il Castelletto" (Trieste)

2022 Communicating AI, and communicating with AI: challenges and opportunities, Future visions on science communication (Trieste)

- La tua vita in dati, i dati della tua vita, SISSA4Schools (Trieste)
- Dal Big Bang all'Intelligenza Artificiale, Biennale di Tecnologia (Torino)
- Scelte intelligenti? Come l'intelligenza artificiale entra nella nostra vita, CICAP Fest (Padova)
- G-astronomy: A culinary and multi-sensorial exploration of the Universe (online)

2021 Il cielo di domani, fra satelliti e stelle, workshop al Convegno Nazionale di Comunicazione della Scienza (Trieste)

- "Stelle Scadenti", Scienza e Virgola science festival, Trieste
- Understanding the Universe with AI, Loughton Astronomical Society (online)
- Cosmo e dati, panel discussion for SISSA Schools open day (online)
- Understanding the Universe with AI, St John's College Oxford (online)

2020 From the Big Bang to AI, Imperial Science Breaks, online

- Big Bang Data, ESOF 2020, Trieste
- SISSA Summer Science Festival, Miramare Castle, Trieste
- Language of the stars, super/collider (online)
- The Solar System for Kids, Building Blocks Nursery, Wimbledon, London
- G-Astronomy Cosmic Dinner, Uig Sands restaurants, Uig bay
- G-Astronomy Cosmic Cocktails and Canapés, An Lanntair restaurant, Stornoway
- From the Big Bang to AI, Professorial Inaugural Lecture, Imperial College London

2019 Keynote, Web Summit DeepTech Stage, Lisbon

- Liceo Cantonale di Locarno, Talk for the general public, Switzerland
- Liceo Cantonale di Locarno, Talk to high school students, Switzerland
- Herstmonceaux Science Festival
- Spotlight talk, Great Exhibition Road Festival, London
- Multi-Sensorial Dark Matter Experience, Great Exhibition Road Festival, London
- Poplar Primary School Space Day (main organizer), Merton Park, London
- Multi-sensorial Dark Matter Experience, Science Museum Lates, London

2018 New Scientist Live, London

- Dining with Copernicus, immersive dining experience, Ognisko Restaurant/Polish Cultural Institute, London
- STEM workshop for high ability students with high-functioning autism/Asperger's, Imperial College London
- Public Engagement Day Pecha Kucha, Imperial College London
- The Science of Star Trek, Royal Albert Hall
- Stephen Hawking's Universe explained in 1,000 simple words, El Debats del Magnànim, Valencia
- Craters on the Moon, Early Years Education Centre, Imperial College London
- The Alternative Guide to STEM, Queen's Park Community School, London, winner of audience vote
- The Map of the Universe, Data Science Institute, Imperial Festival
- The Science of Star Trek, Imperial Festival

2017 EdFoo Camp, Google HQ, Mountain View

- Il Tempo nell'Universo, Università della Svizzera Italiana, Lugano, Switzerland
- g-ASTRONOMY for people with visual impairment, London
- JV Parekh International School, Mumbai, India
- Science City Auditorium, Ahmedabad, India
- IUCAA, Pune, India
- Birla Planetarium, Chennai, India
- Second Home cultural programme, London
- CLCC Research Seminar, Imperial College London
- Being Human Festival, London

2016 g-ASTRONOMY, Cheltenham Science Festival

- Panel: To Infinity and Beyond, Imperial Festival, Imperial College London
- g-ASTRONOMY, Imperial Festival, Imperial College London
- Early Years Education Centre, Imperial College London

- Jersey School for Girls, Jersey
- St Mary's Primary School, Jersey
- Jersey Astronomy Club, Jersey
- Why Society Needs Astronomy and Cosmology, Guest Gresham College Lecture, Museum of London
- A Matter of Gravity, Royal Astronomical Society public lecture
- Big Bang! Daresbury Laboratory

2015 g-ASTRONOMY: the cosmos at the tip of your tongue, Science Museum Lates, London

- Invited talk, Guadalajara International Book Fair, Mexico
- Panel: Science in the UK, Guadalajara International Book Fair, Mexico
- Panel: The Language of Science, Guadalajara International Book Fair, Mexico
- Exploring the Universe, Guadalajara International Book Fair, Mexico
- Panel: What scientists can gain from literature, Guadalajara International Book Fair, Mexico
- International Astronomy Colloquium Round table, Guadalajara International Book Fair, Mexico
- Creative Quarter invited talk, London
- The Physics of Star Trek, Royal Albert Hall, London
- The Blackett Colloquium, Imperial College London
- Imperial Horizons guest lecture, Imperial College London
- Across RCA workshop, Imperial College London
- Interactive cosmology workshop, Technopop Brixton
- Invited talk, Science For Fiction 2015, Imperial College London
- Imperial College London Open Day talk
- Invited lecture, Graveney School Physics Society
- Lightning talk, Google SciFoo Camp, Mountain View
- Keynote speech, Graduate School Symposium, Imperial College London
- Imperial Open Day talk, Imperial College London
- A-level school visit, Imperial College London
- Book tour talk, Zurich Dalkey Book Festival, Dublin
- The Great Cosmic Cookery Show, The Time Cheltenham Science Festival
- Performance at art show opening, Institute for Contemporary Art, London
- Book tour event, Imperial Festival
- Big Bang Bash event, Edinburgh International Science Festival
- Book tour talk, Edinburgh International Science Festival
- Soapbox talk, Edinburgh International Science Festival
- Invited lecture, Airware, San Francisco
- Invited talk, Royal Institution
- Book tour talk, Imperial Fringe Festival
- Invited talks, Winchester planetarium

2014 Book tour talk, Commonwealth Club of California, San Francisco

- The Edge of the Sky, Authors at Google, Mountain View
- Book tour talk, Town Hall Seattle, Seattle
- Book tour talk, Powell's Bookstore, Portland, OR
- Public lecture, Technopop, Stratford London
- Invited lecture, Manchester Science Festival, Manchester
- Invited lecture, World Teach In weekend, London
- Invited lecture, English Language Council lecture, London
- Multiple public lectures, Imperial Fringe Festival, London
- Exhibitor, World Science Fiction Convention 2014, London
- IOP public lecture, Salisbury
- Public lecture, Imperial College London
- Great Cosmic Cookery Show, Imperial Festival
- Invited talk, The Observatory, Hertsmonceaux astronomy festival
- Invited talk, Science for Fiction workshop, London
- Science workshop for families, Cosmic Fun Club, Glasgow Science Centre

- Public talk, The Great Cosmic Cookery Show, Imperial Festival, London
- Science workshop for young children, MoonBerry Muffins!, Winchmore Hill
- Our place in the multiverse, TEDxHackney 'Radical Beauty'
- Invited lecture, Alleyn's School, London
- Invited lecture, St Paul's Way Trust School, Tower Hamlet
- Multiple invited classroom activities, Norland Place School, London
- Classroom activity, St. Joseph's Primary School, London
- Invited talk, Bishop Challoner Catholic Collegiate School, London

2013 Invited lecture, The London Oratory School, London
- Invited talk, Wimbledon High School, Wimbledon
- Invited talks, MENSA Society, Birmingham
- Public talk, SPACE at the White Building, London

2012 Invited talk, AstroSoc, Imperial College London
- Public talk, Università della Svizzera Italiana, Lugano, Switzerland

2011 Invited talk, Friends of the RAS, London
- Public talk, Crayford Manor astronomical society

2010 Invited talk, Westminster School, London

2009 Invited talk, Marlborough College
- Public talk, HGS Astronomical Society
- Public talk, Newbury Astronomical Society, Newbury
- Multiple invited lectures, Festival de Fleurance, France
- Public talk, Centro Stefano Franscini, Ascona, Switzerland
- Public talk, INTECH Planetarium, Winchester
- Public talk, Daresbury Laboratory, Warrington
- Multiple invited lectures, Kassel High School, Kassel, Germany
- Exhibitor, Royal Society Summer Science Exhibit
- Invited lecture, AA School of Architecture, London

2008 Invited talk, Association for Astronomy Education's Annual General Meeting, London
- Public talk and panel discussion, EuroScience Open Forum, Barcelona
- Family event, Meet the Scientist! Museum of Science and Industry, Manchester
- Invited talk, Tudor Hall School, Banbury
- Public lecture, Winchester Planetarium
- Public talk, CornerClub, Oxford
- Invited talk, Green College, Oxford
- Invited lecture, National Particle Physics Masterclass at Oxford University
- Multiple lectures, Queen Mary 2 Caribbean Adventure voyage
- Public lecture, Oxford University Astrophysics Department

2007 Invited lecture, Oxford University Space and Astronomy Society, Oxford
- Invited talk, Bedford school, Bedford
- The BA Award Lectures: Lord Kelvin Lecture 2007, York
- Public lecture, Royal Astronomical Society, London
- Invited lecture, Marlborough College Summer School

2006 Lead exhibitor, Royal Society Summer Science Exhibit, London
- Invited activity, Gonville Primary School
- Invited lecture, Mansfield College, Oxford
- Invited activity, Sonning Common primary school

2005 Invited talk, Bedford school
- Public lecture, CERN, Geneva
- Invited lecture, Astronomy club of Divonne, France

2004 Public talk, CERN, Geneva

Science Communication Teaching and Training

2025 Invited panelist, 'Giornalismo ed intelligenza artificiale', training day for professional journalists, Trieste

2024 Lecturer, 'Intelligenza Artificiale: Guida al Cambiamento', training programme for school teachers, Gradisca

2020 Lecturer, 'Comunicare la ricerca', Science Communication Masters programme, SISSA

2018 Workshop leader, 'The Sound of Space', Imperial College London and Royal Albert Hall
- Public Engagement Academy Panel member, Imperial College London
- Public Engagement for Research Fellows, Imperial College London
- Communicating Science guest lecture MSc programme, Imperial College London

2017 Science Communication Lecture, ISAPP Summer School
- Communicating Science guest lecture, Imperial College London

Courses for the General Public

2021–22 Gresham College Visiting Professor of Cosmology: "The frontiers of knowledge" (online and in person): The Future of Life on Earth
- The Broken Cosmic Distance Ladder
- Einstein's blunder

2020–21 Gresham College Visiting Professor of Cosmology: "The unexpected Universe" (online): Space sounds: the music of the cosmos
- Neutrino: the particle that shouldn't exist
- Understanding the Universe with AI

2019–20 Gresham College Visiting Professor of Cosmology: "The nature of reality" (Museum of London): Mysteries of the dark cosmos
- What has Einstein ever done for you?
- Weighing the Universe

2014 Invited lecture, 36th Annual Astronomy Weekend, Oxford

2010 Day course (with Matthew Malek), Oxford University Continuing Education

2009 Day course (with Bob Lambourne), Oxford University Continuing Education
- Week-long cosmology course, Oxford University Summer School for Adults
- Multiple lectures, Oxford University Summer School

2008 Weekend course, multiple lectures, 30th Annual Astronomy Weekend
- 20-lectures series on cosmology, Oxford University Continuing Education department
- Day course (with Bob Lambourne), Oxford University Continuing Education department

2007 Day course (with Bernard Carr), Oxford University Continuing Education department

2006 20-lectures series on cosmology, Oxford University Continuing Education department

Art / Science and Data Journalism Projects

2026 Scientific advisor, *La storia dell'Universo*, permanent exhibit, SISSA

2026 Scientific advisor, *The unwetlands*, A Satellite-Based Investigation into Italy's Lost Wetlands, with FACTA.eu (Elisabetta Tola and others)

2023 Producer and scientific advisor, *Parl-IA-moci*, an original theatre play by and with Diana Höbel, featuring an impossible dialogue between Alma Mahler (played by Diana) and Gustav Mahler (played by ChatGPT). AI consultant. Premiered at the Modena Festival di Filosofia 2023 (Sept 2023)
- Soroptimist Club, Trieste (May 2024)
- Scienza & Virgola Festival, Trieste (May 2024)
- Invisible Cities, Gorizia (Aug 2024)
- Teatro Rossetti, Trieste (Apr)

2023 Scientific advisor, *Five centuries in five minutes*, in collaboration with Gigi Funcis, an AI-generated video for the exhibit *Kosmos*, Miramare Castle, Trieste

2022 Co-author and creative advisor., *The Edge of the Sky / Oir Nan Speur: A theatre show in Gaelic and English*, by production company sruth-mara. Premiere in 2022 at the Dark Sky Festival, Stornoway, Scotland

2021–22 Featured in *Ophelia in Exile* by Dr Tereza Stehlikova, multimedia art installation, Czech Centre London

2021	Artistic director, concept and script, *LIBRA*, an original multimedia theatre play (with director and video-artist Gigi Funcis), commissioned by SISSA Summer Festival (2021). Nationwide media coverage, including *Il Messaggero, Il Corriere della Sera, Radio Rai 1 FVG, Sky TG24, Il Sole 24 ore, Il Piccolo, RAI Radio 3, Rete4 local TV, etc.* Performed at Miramare Castle,Trieste (Sept 2021; Sept 2022), Gradisca (Oct), Reggio Calabria (opening of Cosmos Festival, Oct 2022), Gorizia (part of GO25!, European Capital of Culture, 2025)
	▹ Featured in *SISSA hosting*, an immersive audio installation part of the *Invisible Cities Festival 2021*
2019	Lead designer, A Multi-Sensorial Dark Matter Experience: in collaboration with human-machine interaction experts, a voyage for two using all five senses to 'perceive' dark matter, Great Exhibition Road Festival, London
2018	*Dinner with Copernicus*, immersive dining experience (with Lab Collective, The Polish Cultural Institute and Ognisko Restaurant), London
2017	Lead designer, *g-ASTRONOMY*: in collaboration with molecular gastronomy chef Josez Youssef and the Royal National Institute for the Blind, a tactile, olfactory and edible experience illustrating cosmological, London
2015	Co-author, *All There Was* (with artists O. Hagen and D. Cheeseman), art show at the Institute for Contemporary Art, part of *fig-2*
2014	Featured in *The All-There-Is*, short documentary by filmmaker L. Melesio Friedman, winner of "Best Portrait of the Life of a Physicist" award of the FQXi Video Competition (2014), `tinyurl.com/o6edrms`
2013	Scientific advisor, *The Theory of Everything* short film, winner of the audience award, Viral Video Awards at the 29th International Short Film Festival Berlin (2013)
	▹ Co-proposer, *The Post-Newtonian Orrery* (with artists O. Hagen and D. Cheesman), 100 best Artangel proposals in
2011–2012	Scientific advisor, *Urban Sputnik* (by designer V. Harden), exhibited at the Royal Institution, Royal Astronomical Society, Imperial College and Kinetica art fair (2011–2012)
2009–2011	Co-author, *Beyond Entropy* research programme (with architect J. Löffler and artist P. Liversidge), exhibited at the Venice Architecture Biennale, the Milano Triennale and the London Architectural Association (2009–2011)

Other Activities

Jury member, *Premio Cosmos*, Reggio Calabria (2022–); International FameLab semi-final judge, Cheltenham (2015); Project advisory board, 'Dark Matters: an interrogation of thresholds of (im)perceptibility through theoretical cosmology, fine art and anthropology of science', U. Lancaster (2014–2016); Coordinator, Astrophysics Group outreach activities, Imperial College London (2008–2015).

Broadcast Media Appearances

2026	'Terra chiama terra' RAI FVG, Apr 28th
	▹ Countercurrent podcast, with Roger Kneebone, Jan 12th
2025	'Millevoci' solo interview, RSI Rete 1, Dec 10th 2025
	▹ 'What a wonderful world' RAI FVG Radio, Nov 20th
	▹ Retesette TV, May 15th
	▹ *Il Giardino di Albert*, Swiss National TV, Mar 14th, 2025
	▹ *Radio3 Scienza*, RAI National Radio, March 10th
2024	*The Infinite Monkey Cage — Starless Worlds*, BBC Radio 4, Nov
	▹ *How to Academy — How the stars made us,* with Robin Ince, Nov
	▹ RAI RadioScienza, Oct 29th
	▹ *Calanais Conversations*, Oct
	▹ Aula di Scienze: Turing e l'AI nella ricerca scientifica, May 30th
	▹ Solar Eclipse Magic, in: *Living on Earth*, March 15th 2024
	▹ Buongiorno Regione, RAI FVG, Feb 20th
2023	Start of the Week, BBC Radio 4, Nov
	▹ Economy.bg, Nov 2023
	▹ KPCW-Cool Science Radio, Nov 2023
	▹ Nova TV (Bulgaria), Nov 2023
	▹ Bulgaria National TV, Nov

- ReggioTV, Telegiornale, Oct 7th
- *Millevoci*, Swiss National Radio, Rete 1 Sept 26th 2023
- TGR Friuli Venezia-Giulia, Sept 23rd
- Sei di sera, RSI Rete 1, Aug 23rd
- Radio 3 Scienza, Italian National radio, May
- In conversation with with Melanie Mitchell, Premio Cosmos Studenti, Apr 2023 (online)

2022 *Telegiornale delle 12.30*, RSI Swiss National TV, Oct 12th
- Sei di sera, RSI Rete 1, July 12th
- In conversation with Jimena Canales, Premio Cosmos Studenti, Apr 2022 (online)
- Hebridean Dark Skies Festival podcast, Jan 21st

2021 *Sky TG24*, Sept 9th 2021
- Radio Capodistria, Sept 9th 2021
- *Radio3 Scienza*, Italian national radio, Sept 6th 2021
- Live interview, RTL 102.5 radio, Sept 5th 2021
- Live appearance, *Tele4* local TV, Aug 31st 2021
- *RAI FVG Giornale Radio*, May 28th 2021

2020 *Beyond the Fourth Floor*, Imperial College students podcast, online
- *Retequattro Estate*, Local TV interview, Trieste, Italy

2019 *Millevoci: Solo interview*, Swiss National Radio, Rete 1
- *Millevoci: Buchi Neri*, Swiss National Radio, Rete 1

2018 *Millevoci: Stephen Hawking*, Swiss National Radio, Rete 1
- *Telegiornale Edizione delle 20*, RSI Swiss National TV
- *Millevoci: Hubble e le prime stelle*, Swiss National Radio, Rete 1
- *Millevoci: Da dove nasce il tempo?*, Swiss National Radio, Rete 1

2017 *Millevoci: Onde Gravitazionali*, Swiss National Radio
- *Telegiornale Edizione delle 20*, RSI Swiss National TV
- *Science Mixtape*, Soho Radio
- *Chiacchiere Cosmologiche*, Swiss National Radio, Rete 3
- *Millevoci*, Swiss National Radio, Rete 1

2016 *Mission Impossible*, ICradio show
- *Albachiara*, Radio Svizzera Rete 1
- *Millevoci*, Swiss National Radio, Rete 1

2015 *Einstein's Extraordinary Universe*, ABCcatalyst Australia
- *Neil Delamere's Sunday Best*, Today FM, Dublin
- *The Sky at Night*, BBC 4
- *Sierra Club Radio* with Orli Cotel

2014 *Albachiara*, Radio Svizzera Rete 1
- *Al Jazeera Arabic*, news report, Nov
- *Inquiry*, WICN public radio, with Mark Lynch
- *Between the Covers*, KBOO FM, with Leigh Anne Kranz
- *Living on Earth*, with Steve Curwood
- *Words on a Wire*, KETP radio, with Daniel Chacòn
- *The Jason Mohammad Show*, BBC Radio Wales
- *Talk Radio Europe*, with Hannah Murray
- *C-SPAN2 Book TV*, talk at the Commonwealth Club of California
- *KATU-TV am northwest*, with Helen Raptis and Dave Anderson
- *NPR's Weekend Edition Sunday*, with Wade Goodwyn

2010 *Baobab*, Swiss National Radio Rete 3

2009 *Il Giardino di Albert*, Swiss National TV RSI
- *Millevoci*, Swiss National Radio, Rete 1

2008 *Future Generation Thinkers*, BBC Radio 3

Coverage in Print and Online

2026 *Monocle*, July/Aug 2026

2025 *universi* INAF magazine, Dec 2025

- *War Room InnovAction*, with Luca de Biase, Apr 2025
- *Restorations Conversations* magazine, Issue 7, Spring 2025
- Scienzainrete interview, Mar 21st 2025
- *RSI Info* website, Mar 16th 2025
- *Il Piccolo* newspaper, Jan 10th

2024 Meridiana 291, Sett-Ott 2024, 12–15

- Let's Talk: Learning About AI Through Art, Aspen Institute blog, Oct 25h
- E se non ci fossero le stelle?, Mangrovia magazine, Sept 9th

2022 The Naked Scientist: Can AI help us understand dark energy better?, podcast, Sept 30th

- *LeScienze*, March 16th

2021 *Il Piccolo* newspaper, Apr 28th

- *Dalla Cosmologia al Razzismo*, con Ruggero Rollini, online video

2020 *Il Piccolo* newspaper, Oct 27th

2018 *CrowdScience, Why Does Dark Matter, Matter?*, BBC World Service, podcast

2017 *Countercurrent* podcast, with Roger Kneebone

- *Scientists, not the Science* podcast, with Stuart Higgins

2016 *Institute of Physics* podcast, with James Dacey

- *Jersey Evening Post* feature article, Apr 23rd
- *reporter* Imperial College London magazine interview
- *A New Language For Science*, Science Writing Resources for Learning blog

2015 *The Reading Survey*, Edinburgh Science Festival

- *Fascination* magazine interview, Summer 2015
- fig. 2 "All There Was" podcast, with Jessica Temple
- *FQXi* podcast, with Sophie Hebden
- *Fiat Physica* online interview
- *Book Q&A*, with Deborah Kalb

2014 *The Guardian Astronomy Science Weekly* podcast, with Nicola Davies

- *Institute of Astrophysics of the Canary Islands* podcast
- *Authors at Google*, video of talk at Google HQ
- *Il Caffè*, Swiss Italian newspaper interview
- *popularscience.co.uk* blog post
- *Science for the People* podcast, with Desiree Schell
- *Speaking of Science* podcast, with Julie Gould
- *Can the universe be described in simple English?* British Council's VOICES blog
- *Living on Earth* podcast, with Steve Curwood
- *New Books in Physics* podcast, with Meg Rosenburg
- *In Residence* podcast, with Steve Scher
- *Imperial Podcast*, with Gail Wilson
- *Nature* podcast, with Elizabeth Gibney

2013 *SCOPE* magazine interview

2012 *Pod Academy* podcast

- *IOP 100-seconds physics* videos